

Indexing

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

→ 2 1

3D
array

	0	1
0	[1 2 3 4]	
1	[5 6 7 8]	
2	[9 10 11 12]	

depth row column
[2, 0, 1]

Slicing

→ [start : end : step-size]

↓ ↓ ↓
0 3 1

0	1	2	3	4	5	6	7
1	2	3	4	5	6	7	8

→ [3 4]

arr_2d →

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

arr_2d[2:4, 2:4]

arr_2d [1:3 , 1:3]

↑ exclusive

row column

Dimension 1 → contains 3 matrices

Dimension 2 → Each matrix has
2 rows

0 1 2

[[[1 2 3]
[4 5 6]]

[[7 8 9]
[10 11 12]]

[[13 14 15]
[16 17 18]]

Dimension 3 → Each matrix has 3 columns

1	2	3
7	8	9
13	14	15

arr_3d [:, 0, ...]

first row in the second dimension

all matrices

all remaining dimensions

arr_3d[:, 0, 0:2]

0, 1 column
↓
oth row

include all the

submatrices

[[[1 2 3]
[4 5 6]]

[[7 8 9]
[10 11 12]]

[[13 14 15]
[16 17 18]]

	0	1	2
0	1	2	3
1	7	8	9
2	13	14	15

	1	2
0	2	3
1	8	9
2	14	15

arr2 = np.array([[5, 2, 3],
[8, 1, 6],
[7, 4, 9]])

	0	1	2
0	5	2	3
1	8	1	6
2	7	4	9

axis = 0

axis = 1

2	3	5
1	6	8
4	7	9

5	1	3
7	2	6
8	4	9

size of an array
split count

→ properly
divisible

✓
equal count of
elements in
each array

Identity Matrix

1	0	0
0	1	0
0	0	1